

```
In [1]: import pandas as pd
df=pd.read_csv("Encoding Data.csv")
df
```

```
Out[1]:
```

	id	bin_1	bin_2	nom_0	ord_2
0	0	F	N	Red	Hot
1	1	F	Y	Blue	Warm
2	2	F	N	Blue	Cold
3	3	F	N	Green	Warm
4	4	T	N	Red	Cold
5	5	T	N	Green	Hot
6	6	F	N	Red	Cold
7	7	T	N	Red	Cold
8	8	F	N	Blue	Warm
9	9	F	Y	Red	Hot

```
In [2]: from sklearn.preprocessing import LabelEncoder,OrdinalEncoder
pm=['Hot','Warm','Cold']
e1=OrdinalEncoder(categories=[pm])
e1.fit_transform(df[["ord_2"]])
```

```
Out[2]: array([[0.],
               [1.],
               [2.],
               [1.],
               [2.],
               [0.],
               [2.],
               [2.],
               [1.],
               [0.]])
```

```
In [3]: df['bo2']=e1.fit_transform(df[["ord_2"]])
df
```

```
Out[3]:
```

	id	bin_1	bin_2	nom_0	ord_2	bo2
0	0	F	N	Red	Hot	0.0
1	1	F	Y	Blue	Warm	1.0
2	2	F	N	Blue	Cold	2.0
3	3	F	N	Green	Warm	1.0
4	4	T	N	Red	Cold	2.0
5	5	T	N	Green	Hot	0.0
6	6	F	N	Red	Cold	2.0
7	7	T	N	Red	Cold	2.0
8	8	F	N	Blue	Warm	1.0
9	9	F	Y	Red	Hot	0.0

```
In [4]: le=LabelEncoder()
dfc=df.copy()
dfc['ord_2']=le.fit_transform(dfc['ord_2'])
dfc
```

```
Out[4]:
```

	id	bin_1	bin_2	nom_0	ord_2	bo2
0	0	F	N	Red	1	0.0
1	1	F	Y	Blue	2	1.0
2	2	F	N	Blue	0	2.0
3	3	F	N	Green	2	1.0
4	4	T	N	Red	0	2.0
5	5	T	N	Green	1	0.0
6	6	F	N	Red	0	2.0
7	7	T	N	Red	0	2.0
8	8	F	N	Blue	2	1.0
9	9	F	Y	Red	1	0.0

```
In [1]: from sklearn.preprocessing import OneHotEncoder
ohe=OneHotEncoder(sparse=False)
df2=df.copy()
enc=pd.DataFrame(ohe.fit_transform(df2[['nom_0']]))
```

NameError Traceback (most recent call last)

Cell In[1], line 3

```
1 from sklearn.preprocessing import OneHotEncoder
2 ohe=OneHotEncoder(sparse=False)
----> 3 df2=df.copy()
      4 enc=pd.DataFrame(ohe.fit_transform(df2[['nom_0']]))
```

NameError: name 'df' is not defined

```
In [8]: df2=pd.concat([df2,enc],axis=1)
df2=pd.concat([df2,enc],axis=1)
df2
```

```
Out[8]:
```

	id	bin_1	bin_2	nom_0	ord_2	bo2	0	1	2	0	1	2
0	0	F	N	Red	Hot	0.0	0.0	0.0	1.0	0.0	0.0	1.0
1	1	F	Y	Blue	Warm	1.0	1.0	0.0	0.0	1.0	0.0	0.0
2	2	F	N	Blue	Cold	2.0	1.0	0.0	0.0	1.0	0.0	0.0
3	3	F	N	Green	Warm	1.0	0.0	1.0	0.0	0.0	1.0	0.0
4	4	T	N	Red	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0
5	5	T	N	Green	Hot	0.0	0.0	1.0	0.0	0.0	1.0	0.0
6	6	F	N	Red	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0
7	7	T	N	Red	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0
8	8	F	N	Blue	Warm	1.0	1.0	0.0	0.0	1.0	0.0	0.0
9	9	F	Y	Red	Hot	0.0	0.0	0.0	1.0	0.0	0.0	1.0

```
In [9]: pd.get_dummies(df2,columns=["nom_0"])
```

```
Out[9]:
```

	id	bin_1	bin_2	ord_2	bo2	0	1	2	0	1	2	nom_0_Blue	nom_0_Green	nom_0
0	0	F	N	Hot	0.0	0.0	0.0	1.0	0.0	0.0	1.0	0	0	
1	1	F	Y	Warm	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1	0	
2	2	F	N	Cold	2.0	1.0	0.0	0.0	1.0	0.0	0.0	1	0	
3	3	F	N	Warm	1.0	0.0	1.0	0.0	0.0	1.0	0.0	0	1	
4	4	T	N	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0	0	0	
5	5	T	N	Hot	0.0	0.0	1.0	0.0	0.0	1.0	0.0	0	1	
6	6	F	N	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0	0	0	
7	7	T	N	Cold	2.0	0.0	0.0	1.0	0.0	0.0	1.0	0	0	
8	8	F	N	Warm	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1	0	
9	9	F	Y	Hot	0.0	0.0	0.0	1.0	0.0	0.0	1.0	0	0	

```
In [5]: pip install --upgrade category_encoders
```

```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: category_encoders in c:\users\acer\appdata\roaming\python\python310\site-packages (2.6.3)
Collecting category_encoders
  Using cached category_encoders-2.8.1-py3-none-any.whl (85 kB)
Requirement already satisfied: scipy>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.10.0)
Collecting scikit-learn>=1.6.0
  Using cached scikit_learn-1.7.2-cp310-cp310-win_amd64.whl (8.9 MB)
Requirement already satisfied: pandas>=1.0.5 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.5.3)
Requirement already satisfied: numpy>=1.14.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.23.5)
Requirement already satisfied: patsy>=0.5.1 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (0.5.3)
Requirement already satisfied: statsmodels>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (0.13.5)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.0.5->category_encoders) (2022.7)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\acer\appdata\roaming\python\python310\site-packages (from pandas>=1.0.5->category_encoders) (2.9.0.post0)
Requirement already satisfied: six in c:\programdata\anaconda3\lib\site-packages (from patsy>=0.5.1->category_encoders) (1.16.0)
Requirement already satisfied: joblib>=1.2.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=1.6.0->category_encoders) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=1.6.0->category_encoders) (3.6.0)
Requirement already satisfied: packaging>=21.3 in c:\programdata\anaconda3\lib\site-packages (from statsmodels>=0.9.0->category_encoders) (25.0)
Installing collected packages: scikit-learn, category_encoders
  Attempting uninstall: scikit-learn
    Found existing installation: scikit-learn 1.3.2
    Uninstalling scikit-learn-1.3.2:
      Successfully uninstalled scikit-learn-1.3.2
Note: you may need to restart the kernel to use updated packages.

ERROR: Could not install packages due to an OSError: [WinError 5] Access is denied: 'C:\\Users\\acer\\AppData\\Roaming\\Python\\Python310\\site-packages\\~klearn\\.libs\\msvcp140.dll'
Check the permissions.

```

```
In [11]: from category_encoders import BinaryEncoder
df=pd.read_csv("data.csv")
df
```

```
-----
ImportError                                Traceback (most recent call last)
Cell In[11], line 1
----> 1 from category_encoders import BinaryEncoder
      2 df=pd.read_csv("data.csv")
      3 df

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\__init__.py:9
      1 """Category encoders library.
      2
      3 .. module:: category_encoders
      4 (...)
      5
      6
      7 """
----> 9 from category_encoders.backward_difference import BackwardDifference
Encoder
      10 from category_encoders.basen import BaseNEncoder
      11 from category_encoders.binary import BinaryEncoder

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\backward_difference.py:6
      3 import numpy as np
      4 from patsy.contrasts import ContrastMatrix, Diff
----> 6 from category_encoders.base_contrast_encoder import BaseContrastEncoder
      8 __author__ = 'paulwestenthanner'
      11 class BackwardDifferenceEncoder(BaseContrastEncoder):

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\base_contrast_encoder.py:9
      6 import pandas as pd
      7 from patsy.contrasts import ContrastMatrix
----> 9 import category_encoders.utils as util
      10 from category_encoders.ordinal import OrdinalEncoder
      12 __author__ = 'paulwestenthanner'

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\utils.py:20
      18 from sklearn.exceptions import NotFittedError
      19 from sklearn.preprocessing import LabelEncoder
----> 20 from sklearn.utils import Tags
      22 __author__ = 'willmcginnis'
      24 X_type = np.ndarray | pd.DataFrame | list | np.generic | csr_matrix

ImportError: cannot import name 'Tags' from 'sklearn.utils' (C:\ProgramData\anaconda3\lib\site-packages\sklearn\utils\__init__.py)
```

```
In [6]: be=BinaryEncoder()  
nd=be.fit_transform(df['Ord_2'])  
dfb=pd.concat([df,nd],axis=1)  
dfb1=df.copy()  
dfb
```

```
Out[6]:
```

	id	bin_1	bin_2	City	Ord_1	Ord_2	Target	Ord_2_0	Ord_2_1	Ord_2_2
0	0	F	N	Delhi	Hot	High School	0	0	0	1
1	1	F	Y	Bangalore	Warm	Masters	1	0	1	0
2	2	M	N	Mumbai	Very Hot	Diploma	1	0	1	1
3	3	M	Y	Chennai	Cold	Bachelors	0	1	0	0
4	4	M	Y	Delhi	Cold	Bachelors	1	1	0	0
5	5	F	N	Delhi	Very Hot	Masters	0	0	1	0
6	6	M	N	Chennai	Warm	PhD	1	1	0	1
7	7	F	N	Chennai	Hot	High School	1	0	0	1
8	8	M	N	Delhi	Very Hot	High School	0	0	0	1
9	9	F	Y	Delhi	Warm	PhD	0	1	0	1

```
In [13]: !pip install category_encoders

from category_encoders import BinaryEncoder
import pandas as pd

be = BinaryEncoder()

nd = be.fit_transform(df['Ord_2'])

dfb = pd.concat([df, nd], axis=1)

dfb1 = df.copy()

dfb
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: category_encoders in c:\users\acer\appdata\roaming\python\python310\site-packages (2.8.1)

Requirement already satisfied: patsy>=0.5.1 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (0.5.3)

Requirement already satisfied: scikit-learn>=1.6.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from category_encoders) (1.7.2)

Requirement already satisfied: pandas>=1.0.5 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.5.3)

Requirement already satisfied: numpy>=1.14.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.23.5)

Requirement already satisfied: statsmodels>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (0.13.5)

Requirement already satisfied: scipy>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders) (1.10.0)

Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.0.5->category_encoders) (2022.7)

Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\acer\appdata\roaming\python\python310\site-packages (from pandas>=1.0.5->category_encoders) (2.9.0.post0)

Requirement already satisfied: six in c:\programdata\anaconda3\lib\site-packages (from patsy>=0.5.1->category_encoders) (1.16.0)

Requirement already satisfied: joblib>=1.2.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=1.6.0->category_encoders) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=1.6.0->category_encoders) (3.6.0)

Requirement already satisfied: packaging>=21.3 in c:\programdata\anaconda3\lib\site-packages (from statsmodels>=0.9.0->category_encoders) (25.0)

```

-----
ImportError                                Traceback (most recent call last)
Cell In[13], line 3
      1 get_ipython().system('pip install category_encoders')
----> 3 from category_encoders import BinaryEncoder
      4 import pandas as pd
      6 be = BinaryEncoder()

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\__init__.py:9
      1 """Category encoders library.
      2
      3 .. module:: category_encoders
      4 (...)
      5
      6
      7 """
----> 9 from category_encoders.backward_difference import BackwardDifference
Encoder
      10 from category_encoders.basen import BaseNEncoder
      11 from category_encoders.binary import BinaryEncoder

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\backward_difference.py:6
      3 import numpy as np
      4 from patsy.contrasts import ContrastMatrix, Diff
----> 6 from category_encoders.base_contrast_encoder import BaseContrastEncoder
      8 __author__ = 'paulwestenthanner'
     11 class BackwardDifferenceEncoder(BaseContrastEncoder):

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\base_contrast_encoder.py:9
      6 import pandas as pd
      7 from patsy.contrasts import ContrastMatrix
----> 9 import category_encoders.utils as util
     10 from category_encoders.ordinal import OrdinalEncoder
     12 __author__ = 'paulwestenthanner'

File ~\AppData\Roaming\Python\Python310\site-packages\category_encoders\utils.py:20
     18 from sklearn.exceptions import NotFittedError
     19 from sklearn.preprocessing import LabelEncoder
----> 20 from sklearn.utils import Tags
     22 __author__ = 'willmcginnis'
     24 X_type = np.ndarray | pd.DataFrame | list | np.generic | csr_matrix

ImportError: cannot import name 'Tags' from 'sklearn.utils' (C:\ProgramData\anaconda3\lib\site-packages\sklearn\utils\_init__.py)

```

```
In [7]: !pip uninstall -y scikit-learn category_encoders
```

```

Found existing installation: scikit-learn 1.7.2
Uninstalling scikit-learn-1.7.2:
  Successfully uninstalled scikit-learn-1.7.2
Found existing installation: category-encoders 2.6.3
Uninstalling category-encoders-2.6.3:
  Successfully uninstalled category-encoders-2.6.3

```



```
In [18]: !pip install scikit-learn==1.3.2
!pip install category_encoders==2.6.3
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scikit-learn==1.3.2 in c:\users\acer\appdata\roaming\python\python310\site-packages (1.3.2)
Requirement already satisfied: joblib>=1.1.1 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn==1.3.2) (1.5.3)
Requirement already satisfied: numpy<2.0,>=1.17.3 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn==1.3.2) (1.23.5)
Requirement already satisfied: scipy>=1.5.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn==1.3.2) (1.10.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn==1.3.2) (3.6.0)
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: category_encoders==2.6.3 in c:\users\acer\appdata\roaming\python\python310\site-packages (2.6.3)
Requirement already satisfied: statsmodels>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders==2.6.3) (0.13.5)
Requirement already satisfied: scipy>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders==2.6.3) (1.10.0)
Requirement already satisfied: pandas>=1.0.5 in c:\programdata\anaconda3\lib\site-packages (from category_encoders==2.6.3) (1.5.3)
Requirement already satisfied: numpy>=1.14.0 in c:\programdata\anaconda3\lib\site-packages (from category_encoders==2.6.3) (1.23.5)
Requirement already satisfied: patsy>=0.5.1 in c:\programdata\anaconda3\lib\site-packages (from category_encoders==2.6.3) (0.5.3)
Requirement already satisfied: scikit-learn>=0.20.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from category_encoders==2.6.3) (1.3.2)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.0.5->category_encoders==2.6.3) (2022.7)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\acer\appdata\roaming\python\python310\site-packages (from pandas>=1.0.5->category_encoders==2.6.3) (2.9.0.post0)
Requirement already satisfied: six in c:\programdata\anaconda3\lib\site-packages (from patsy>=0.5.1->category_encoders==2.6.3) (1.16.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=0.20.0->category_encoders==2.6.3) (3.6.0)
Requirement already satisfied: joblib>=1.1.1 in c:\users\acer\appdata\roaming\python\python310\site-packages (from scikit-learn>=0.20.0->category_encoders==2.6.3) (1.5.3)
Requirement already satisfied: packaging>=21.3 in c:\programdata\anaconda3\lib\site-packages (from statsmodels>=0.9.0->category_encoders==2.6.3) (25.0)
```

```
In [8]: from category_encoders import BinaryEncoder
import pandas as pd

be = BinaryEncoder()

nd = be.fit_transform(df['Ord_2'])

dfb = pd.concat([df, nd], axis=1)

dfb
```

```
Out[8]:
```

	id	bin_1	bin_2	City	Ord_1	Ord_2	Target	Ord_2_0	Ord_2_1	Ord_2_2
0	0	F	N	Delhi	Hot	High School	0	0	0	1
1	1	F	Y	Bangalore	Warm	Masters	1	0	1	0
2	2	M	N	Mumbai	Very Hot	Diploma	1	0	1	1
3	3	M	Y	Chennai	Cold	Bachelors	0	1	0	0
4	4	M	Y	Delhi	Cold	Bachelors	1	1	0	0
5	5	F	N	Delhi	Very Hot	Masters	0	0	1	0
6	6	M	N	Chennai	Warm	PhD	1	1	0	1
7	7	F	N	Chennai	Hot	High School	1	0	0	1
8	8	M	N	Delhi	Very Hot	High School	0	0	0	1
9	9	F	Y	Delhi	Warm	PhD	0	1	0	1

```
In [21]: be=BinaryEncoder()
nd=be.fit_transform(df['Ord_2'])
dfb=pd.concat([df,nd],axis=1)
dfb1=df.copy()
dfb
```

```
-----
KeyError                                Traceback (most recent call last)
File C:\ProgramData\anaconda3\lib\site-packages\pandas\core\indexes\base.py:
3802, in Index.get_loc(self, key, method, tolerance)
    3801 try:
-> 3802     return self._engine.get_loc(casted_key)
    3803 except KeyError as err:

File C:\ProgramData\anaconda3\lib\site-packages\pandas\_libs\index.pyx:138,
in pandas._libs.index.IndexEngine.get_loc()

File C:\ProgramData\anaconda3\lib\site-packages\pandas\_libs\index.pyx:165,
in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:5745, in pandas._libs.hashtabl
e.PyObjectHashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:5753, in pandas._libs.hashtabl
e.PyObjectHashTable.get_item()
```

KeyError: 'Ord_2'

The above exception was the direct cause of the following exception:

```
KeyError                                Traceback (most recent call last)
Cell In[21], line 2
      1 be=BinaryEncoder()
----> 2 nd=be.fit_transform(df['Ord_2'])
      3 dfb=pd.concat([df,nd],axis=1)
      4 dfb1=df.copy()

File C:\ProgramData\anaconda3\lib\site-packages\pandas\core\frame.py:3807, i
n DataFrame.__getitem__(self, key)
    3805 if self.columns.nlevels > 1:
    3806     return self._getitem_multilevel(key)
-> 3807 indexer = self.columns.get_loc(key)
    3808 if is_integer(indexer):
    3809     indexer = [indexer]

File C:\ProgramData\anaconda3\lib\site-packages\pandas\core\indexes\base.py:
3804, in Index.get_loc(self, key, method, tolerance)
    3802     return self._engine.get_loc(casted_key)
    3803 except KeyError as err:
-> 3804     raise KeyError(key) from err
    3805 except TypeError:
    3806     # If we have a listlike key, _check_indexing_error will raise
    3807     # InvalidIndexError. Otherwise we fall through and re-raise
    3808     # the TypeError.
    3809     self._check_indexing_error(key)
```

KeyError: 'Ord_2'

```
In [3]: from category_encoders import BinaryEncoder
df=pd.read_csv("data.csv")
df
```

```
Out[3]:
```

	id	bin_1	bin_2	City	Ord_1	Ord_2	Target
0	0	F	N	Delhi	Hot	High School	0
1	1	F	Y	Bangalore	Warm	Masters	1
2	2	M	N	Mumbai	Very Hot	Diploma	1
3	3	M	Y	Chennai	Cold	Bachelors	0
4	4	M	Y	Delhi	Cold	Bachelors	1
5	5	F	N	Delhi	Very Hot	Masters	0
6	6	M	N	Chennai	Warm	PhD	1
7	7	F	N	Chennai	Hot	High School	1
8	8	M	N	Delhi	Very Hot	High School	0
9	9	F	Y	Delhi	Warm	PhD	0

```
In [9]: be=BinaryEncoder()
nd=be.fit_transform(df['Ord_2'])
dfb=pd.concat([df,nd],axis=1)
dfb1=df.copy()
dfb
```

```
Out[9]:
```

	id	bin_1	bin_2	City	Ord_1	Ord_2	Target	Ord_2_0	Ord_2_1	Ord_2_2
0	0	F	N	Delhi	Hot	High School	0	0	0	1
1	1	F	Y	Bangalore	Warm	Masters	1	0	1	0
2	2	M	N	Mumbai	Very Hot	Diploma	1	0	1	1
3	3	M	Y	Chennai	Cold	Bachelors	0	1	0	0
4	4	M	Y	Delhi	Cold	Bachelors	1	1	0	0
5	5	F	N	Delhi	Very Hot	Masters	0	0	1	0
6	6	M	N	Chennai	Warm	PhD	1	1	0	1
7	7	F	N	Chennai	Hot	High School	1	0	0	1
8	8	M	N	Delhi	Very Hot	High School	0	0	0	1
9	9	F	Y	Delhi	Warm	PhD	0	1	0	1

```
In [10]: from category_encoders import TargetEncoder
te=TargetEncoder()
CC=df.copy()
new=te.fit_transform(X=CC["City"],y=CC["Target"])
CC=pd.concat([CC,new],axis=1)
CC
```

```
Out[10]:
```

	id	bin_1	bin_2	City	Ord_1	Ord_2	Target	City
0	0	F	N	Delhi	Hot	High School	0	0.445272
1	1	F	Y	Bangalore	Warm	Masters	1	0.565054
2	2	M	N	Mumbai	Very Hot	Diploma	1	0.565054
3	3	M	Y	Chennai	Cold	Bachelors	0	0.525744
4	4	M	Y	Delhi	Cold	Bachelors	1	0.445272
5	5	F	N	Delhi	Very Hot	Masters	0	0.445272
6	6	M	N	Chennai	Warm	PhD	1	0.525744
7	7	F	N	Chennai	Hot	High School	1	0.525744
8	8	M	N	Delhi	Very Hot	High School	0	0.445272
9	9	F	Y	Delhi	Warm	PhD	0	0.445272

```
In [11]: import pandas as pd
from scipy import stats
import numpy as np
df=pd.read_csv("Data_to_Transform.csv")
df
```

```
Out[11]:
```

	Moderate Positive Skew	Highly Positive Skew	Moderate Negative Skew	Highly Negative Skew
0	0.899990	2.895074	11.180748	9.027485
1	1.113554	2.962385	10.842938	9.009762
2	1.156830	2.966378	10.817934	9.006134
3	1.264131	3.000324	10.764570	9.000125
4	1.323914	3.012109	10.753117	8.981296
...	...	...	...	...
9995	14.749050	16.289513	-2.980821	-3.254882
9996	14.854474	16.396252	-3.147526	-3.772332
9997	15.262103	17.102991	-3.517256	-4.717950
9998	15.269983	17.628467	-4.689833	-5.670496
9999	16.204517	18.052331	-6.335679	-7.036091

10000 rows × 4 columns

```
In [12]: df.skew()
```

```
Out[12]: Moderate Positive Skew    0.656308  
         Highly Positive Skew     1.271249  
         Moderate Negative Skew   -0.690244  
         Highly Negative Skew    -1.201891  
         dtype: float64
```

```
In [13]: np.log(df["Highly Positive Skew"])
```

```
Out[13]: 0      1.063011  
         1      1.085995  
         2      1.087342  
         3      1.098720  
         4      1.102640  
         ...  
         9995    2.790522  
         9996    2.797053  
         9997    2.839253  
         9998    2.869515  
         9999    2.893275  
         Name: Highly Positive Skew, Length: 10000, dtype: float64
```

```
In [14]: np.reciprocal(df["Moderate Positive Skew"])
```

```
Out[14]: 0      1.111123  
         1      0.898026  
         2      0.864431  
         3      0.791057  
         4      0.755336  
         ...  
         9995    0.067801  
         9996    0.067320  
         9997    0.065522  
         9998    0.065488  
         9999    0.061711  
         Name: Moderate Positive Skew, Length: 10000, dtype: float64
```

```
In [15]: np.sqrt(df["Highly Positive Skew"])
```

```
Out[15]: 0      1.701492  
         1      1.721158  
         2      1.722317  
         3      1.732144  
         4      1.735543  
         ...  
         9995    4.036027  
         9996    4.049229  
         9997    4.135576  
         9998    4.198627  
         9999    4.248803  
         Name: Highly Positive Skew, Length: 10000, dtype: float64
```

```
In [16]: np.square(df["Highly Positive Skew"])
```

```
Out[16]: 0      8.381452
1      8.775724
2      8.799396
3      9.001942
4      9.072800
...
9995    265.348230
9996    268.837091
9997    292.512290
9998    310.762852
9999    325.886637
Name: Highly Positive Skew, Length: 10000, dtype: float64
```

```
In [17]: df["Highly Positive Skew_boxcox"],parameters=stats.boxcox(df["Highly Positive S
df
```

```
Out[17]:
```

	Moderate Positive Skew	Highly Positive Skew	Moderate Negative Skew	Highly Negative Skew	Highly Positive Skew_boxcox
0	0.899990	2.895074	11.180748	9.027485	0.812909
1	1.113554	2.962385	10.842938	9.009762	0.825921
2	1.156830	2.966378	10.817934	9.006134	0.826679
3	1.264131	3.000324	10.764570	9.000125	0.833058
4	1.323914	3.012109	10.753117	8.981296	0.835247
...	...	...	...	...	...
9995	14.749050	16.289513	-2.980821	-3.254882	1.457701
9996	14.854474	16.396252	-3.147526	-3.772332	1.459189
9997	15.262103	17.102991	-3.517256	-4.717950	1.468681
9998	15.269983	17.628467	-4.689833	-5.670496	1.475357
9999	16.204517	18.052331	-6.335679	-7.036091	1.480525

10000 rows × 5 columns

```
In [18]: df["Moderate Negative Skew_yeojohnson"],parameters=stats.yeojohnson(df["Modera
df.skew()
```

```
Out[18]: Moderate Positive Skew      0.656308
Highly Positive Skew      1.271249
Moderate Negative Skew    -0.690244
Highly Negative Skew     -1.201891
Highly Positive Skew_boxcox      0.023089
Moderate Negative Skew_yeojohnson -0.119651
dtype: float64
```

```
In [19]: from sklearn.preprocessing import QuantileTransformer
qt=QuantileTransformer(output_distribution='normal')
df["Moderate Negative Skew_1"]=qt.fit_transform(df[["Moderate Negative Skew"]])
df
```

Out[19]:

	Moderate Positive Skew	Highly Positive Skew	Moderate Negative Skew	Highly Negative Skew	Highly Positive Skew_boxcox	Moderate Negative Skew_yeojohnson	Moderate Negative Skew_1
0	0.899990	2.895074	11.180748	9.027485	0.812909	29.137805	5.199338
1	1.113554	2.962385	10.842938	9.009762	0.825921	27.885272	3.227288
2	1.156830	2.966378	10.817934	9.006134	0.826679	27.793301	3.206801
3	1.264131	3.000324	10.764570	9.000125	0.833058	27.597360	3.167111
4	1.323914	3.012109	10.753117	8.981296	0.835247	27.555368	3.159208
...	...	...	...	...	...	...	...
9995	14.749050	16.289513	-2.980821	-3.254882	1.457701	-1.949345	-3.147619
9996	14.854474	16.396252	-3.147526	-3.772332	1.459189	-2.028952	-3.162489
9997	15.262103	17.102991	-3.517256	-4.717950	1.468681	-2.199693	-3.198205
9998	15.269983	17.628467	-4.689833	-5.670496	1.475357	-2.697151	-3.350199
9999	16.204517	18.052331	-6.335679	-7.036091	1.480525	-3.311402	-5.199338

10000 rows × 7 columns

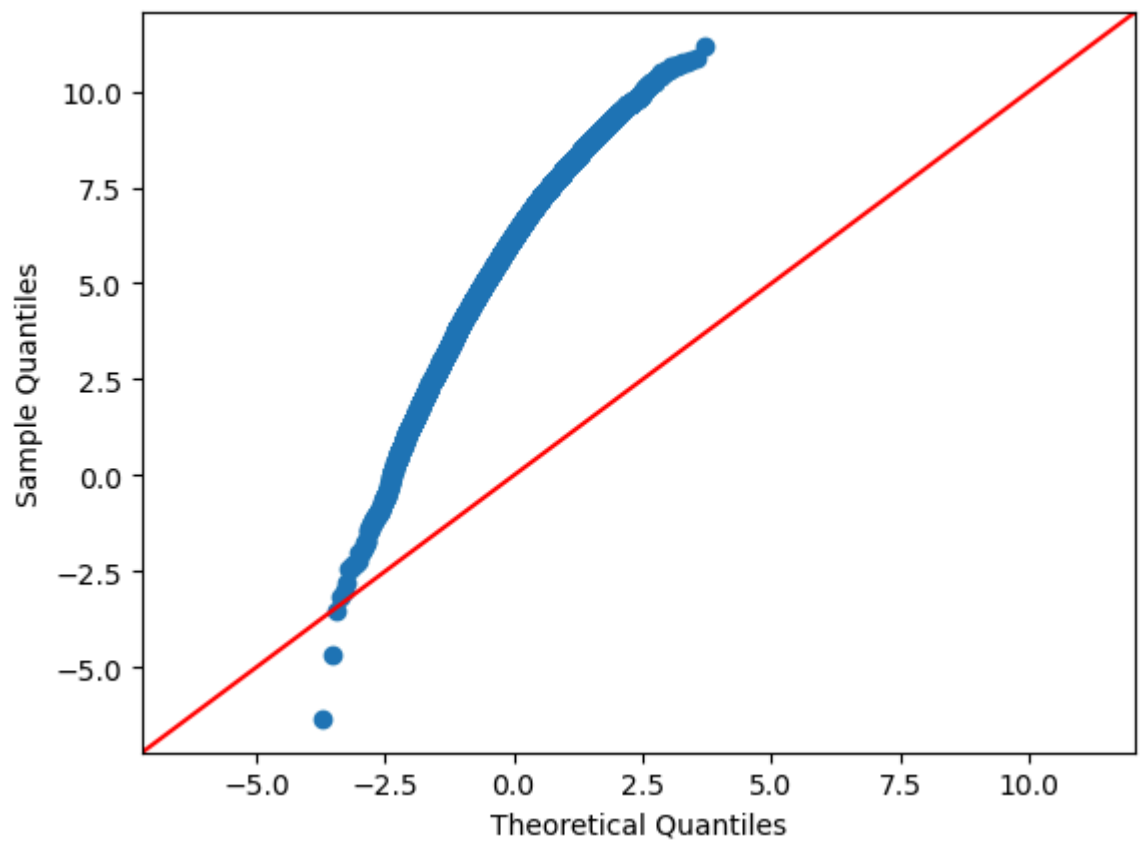
```
In [20]: from sklearn.preprocessing import QuantileTransformer
qt=QuantileTransformer(output_distribution='normal')
df["Moderate Negative Skew_1"]=qt.fit_transform(df[["Moderate Negative Skew"]])
df
```

Out[20]:

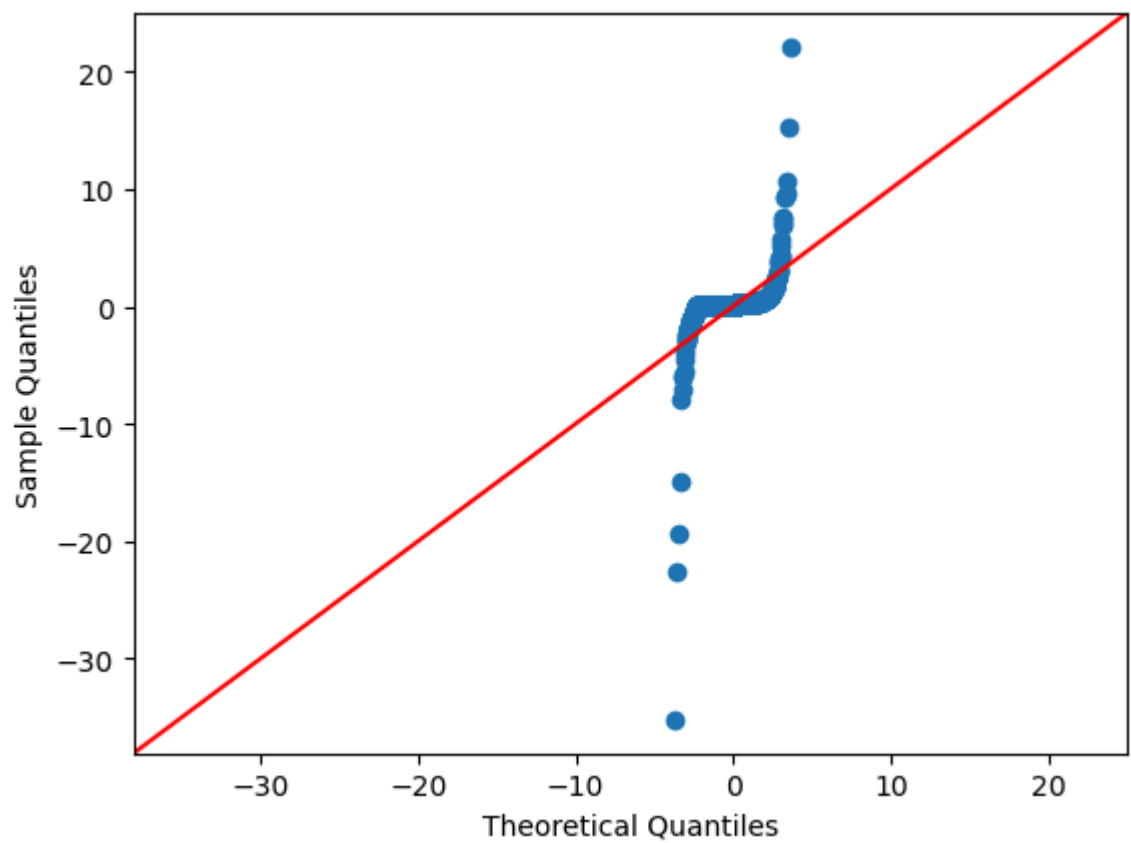
	Moderate Positive Skew	Highly Positive Skew	Moderate Negative Skew	Highly Negative Skew	Highly Positive Skew_boxcox	Moderate Negative Skew_yeojohnson	Moderate Negative Skew_1
0	0.899990	2.895074	11.180748	9.027485	0.812909	29.137805	5.199338
1	1.113554	2.962385	10.842938	9.009762	0.825921	27.885272	3.227288
2	1.156830	2.966378	10.817934	9.006134	0.826679	27.793301	3.206801
3	1.264131	3.000324	10.764570	9.000125	0.833058	27.597360	3.167111
4	1.323914	3.012109	10.753117	8.981296	0.835247	27.555368	3.159208
...	...	...	...	...	...	...	...
9995	14.749050	16.289513	-2.980821	-3.254882	1.457701	-1.949345	-3.147619
9996	14.854474	16.396252	-3.147526	-3.772332	1.459189	-2.028952	-3.162489
9997	15.262103	17.102991	-3.517256	-4.717950	1.468681	-2.199693	-3.198205
9998	15.269983	17.628467	-4.689833	-5.670496	1.475357	-2.697151	-3.350199
9999	16.204517	18.052331	-6.335679	-7.036091	1.480525	-3.311402	-5.199338

10000 rows × 7 columns


```
In [22]: import seaborn as sns
import statsmodels.api as sm
import matplotlib.pyplot as plt
sm.qqplot(df["Moderate Negative Skew"],line='45')
plt.show()
```



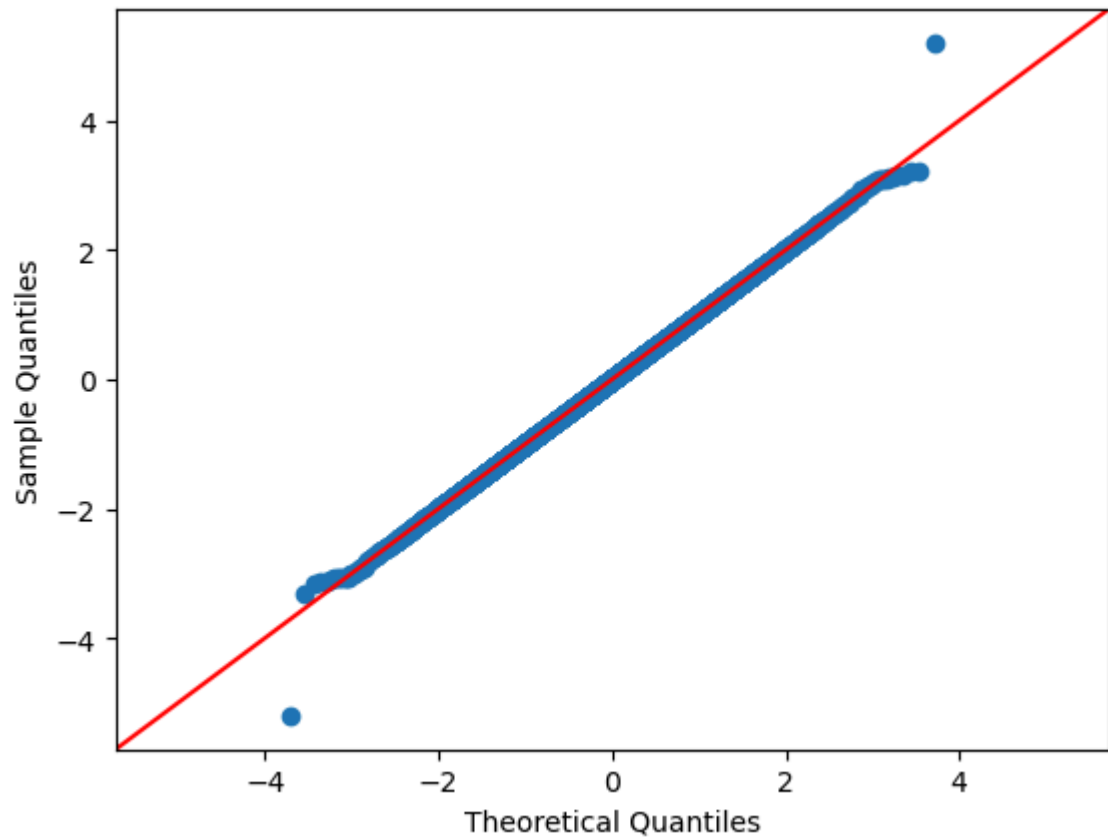
```
In [23]: sm.qqplot(np.reciprocal(df["Moderate Negative Skew"]),line='45')  
plt.show()
```



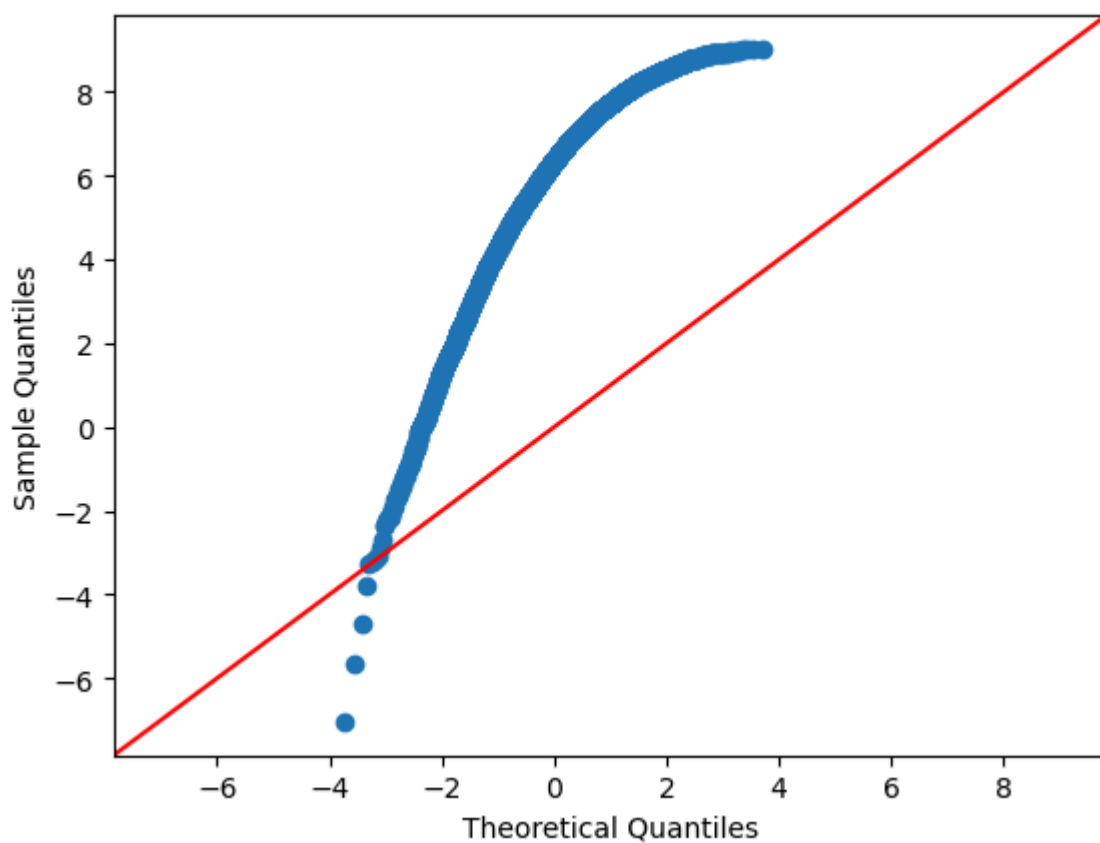
```
In [24]: from sklearn.preprocessing import QuantileTransformer
qt=QuantileTransformer(output_distribution='normal',n_quantiles=891)

df["Moderate Negative Skew"]=qt.fit_transform(df[["Moderate Negative Skew"]])

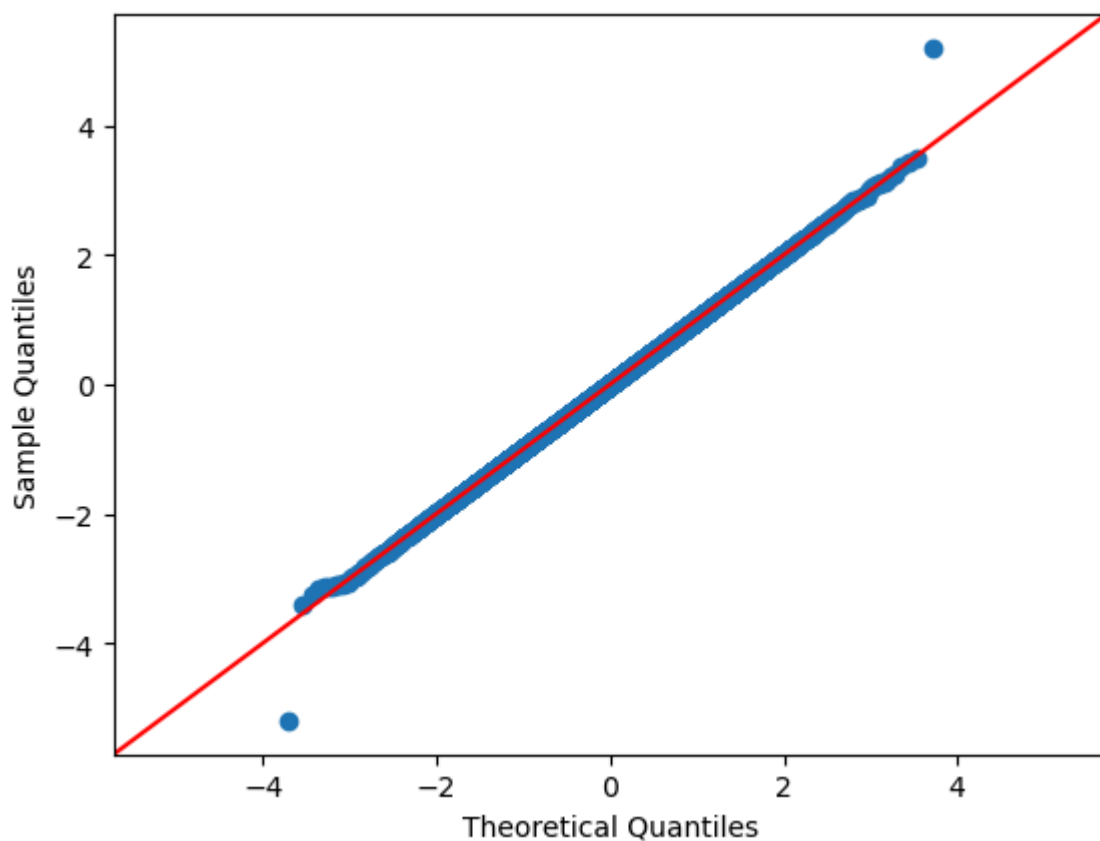
sm.qqplot(df["Moderate Negative Skew"],line='45')
plt.show()
```



```
In [25]: df["Highly Negative Skew_1"]=qt.fit_transform(df[["Highly Negative Skew"]])  
sm.qqplot(df["Highly Negative Skew"],line='45')  
plt.show()
```



```
In [27]: sm.qqplot(df["Highly Negative Skew_1"],line='45')  
plt.show()
```



In []: